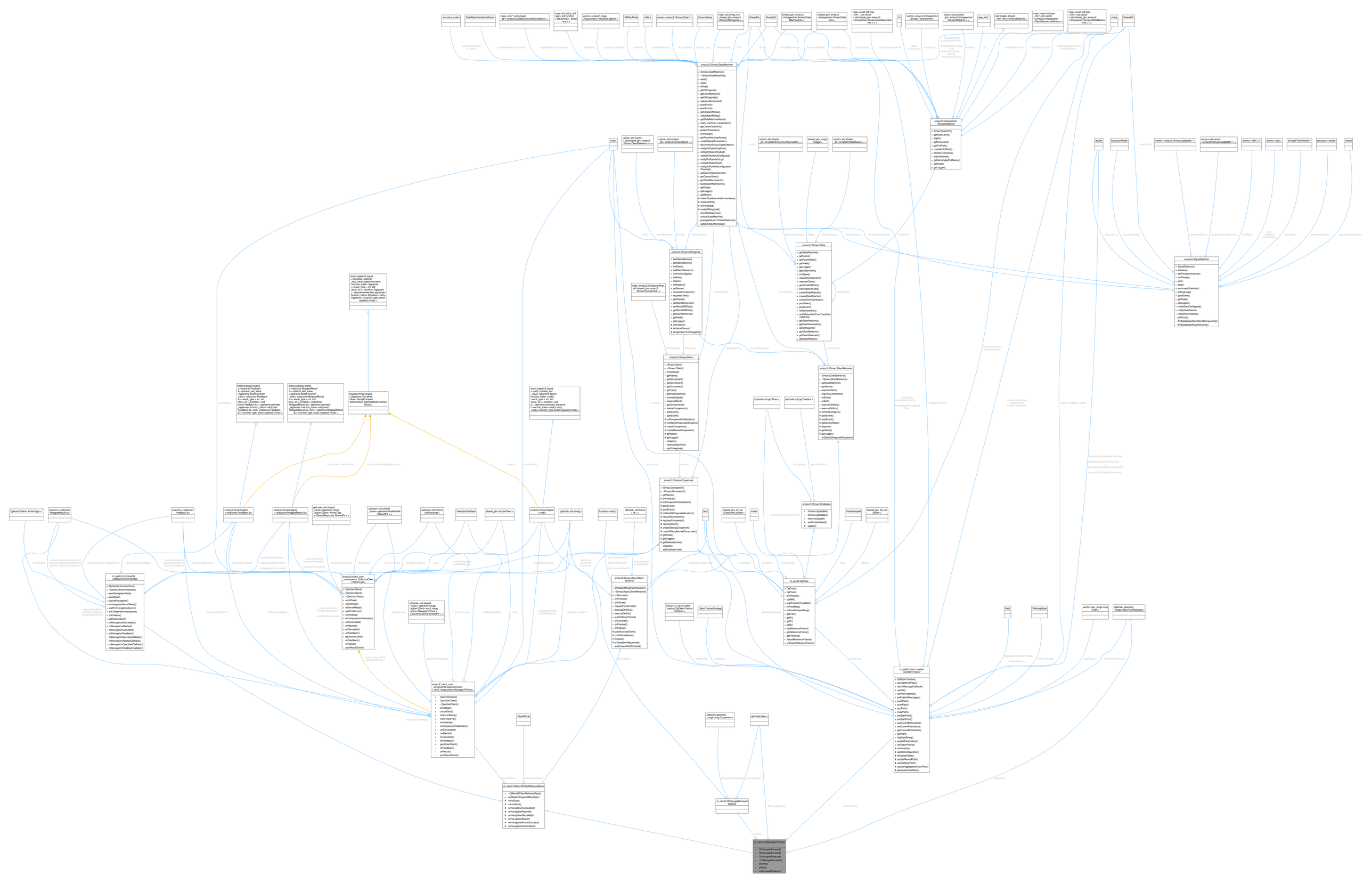

cl_nav2z::CbNavigateForward
+ CbNavigateForward()
+ CbNavigateForward()
+ CbNavigateForward()
+ ~CbNavigateForward()
+ onEntry()
+ onExit()
+ setForwardDistance()
+options
#forwardDistance_
#odomTracker_
#goalPose_
cl_nav2z::CbNavigateForward Options
+forceInitialOrientation
+forwardSpeed
optional< geometry _msgs::msg::Quaternion >
optional< float >
cl_nav2z::CbNav2ZClientBehaviorBase
+ ~CbNav2ZClientBehaviorBase()
+ onStateOrthogonalAllocation()
# sendGoal()
# cancelGoal()
# onNavigationSucceeded()
# onNavigationAborted()
# onNavigationCancelled()
# onNavigationResult()
# onNavigationActionSuccess()
# onNavigationActionAbort()
#actionClient_
#navigationResult_
-goalChecker_
ResultCode
smacc2::client_core _components::CpActionClient < nav2_msgs::action::NavigateToPose >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
< nav2_msgs::action ::NavigateToPose >
#nav2ActionInterface_
-lastCancelResponse_
optional< std::shared _future< typename rclcpp _action::Client< nav2_msgs ::action::NavigateToPose > ::CancelResponse::SharedPtr > >
smacc2::client_core _components::CpActionClient < ActionType >
+ CpActionClient()
+ CpActionClient()
+ ~CpActionClient()
+ sendGoal()
+ cancelGoal()
+ isServerReady()
+ waitForServer()
+ onInitialize()
+ onComponentInitialization()
+ onSucceeded()
+ onAborted()
+ onCancelled()
+ onFeedback()
+ getActionClient()
- onFeedback()
- onResult()
- postResultEvent()
cl_nav2z::components ::CpNav2ActionInterface
+ CpNav2ActionInterface()
+ ~CpNav2ActionInterface()
+ sendNavigationGoal()
+ sendGoal()
+ cancelNavigation()
+ isNavigationServerReady()
+ waitForNavigationServer()
+ onComponentInitialization()
+ onInitialize()
+ getActionClient()
+ onNavigationSucceeded()
+ onNavigationAborted()
+ onNavigationCancelled()
+ onNavigationFeedback()
- onNavigationSuccessCallback()
- onNavigationAbortedCallback()
- onNavigationCancelledCallback()
- onNavigationFeedbackCallback()
CpActionClient< ActionType > *
function< void(const WrappedResult &)>
function< void(const Feedback &)>
smacc2::SmaccSignal < void(const Feedback &)>
smacc2::SmaccSignal < void(const WrappedResult &)>
optional< std::shared _future< typename rclcpp _action::Client< ActionType >::CancelResponse::SharedPtr > >
optional< std::shared _future< typename GoalHandle ::SharedPtr > >
optional< std::chrono ::milliseconds >
FeedbackCallback
shared_ptr< ActionClient >
smacc2::SmaccSignal < void()>
optional< std::string >
function< void()>
optional< std::future < int > >
-actionClient_
+postNavigationAbortedEvent
+postNavigationCancelledEvent
+postNavigationSuccessEvent
+postNavigationFeedbackEvent
+onNavigationFeedback_
+postAbortedEvent
+postCancelledEvent
+postSuccessEvent
+postFeedbackEvent
+onActionFeedback_
+onNavigationAborted_
+onNavigationCancelled_
+onNavigationSucceeded_
+onActionAborted_
+onActionCancelled_
+onActionSucceeded_
-lastCancelResponse_
-lastRequest_
+serverTimeout
-feedbackCallback_
-client_
+onGoalAccepted_
+onGoalRejected_
+actionServerName
-actionMutex_
-onFailure_
-onFinished_
-onSuccess_
-postFailureEventFn_
-postFinishEventFn_
-postSuccessEventFn_
-onEntryThread_
-onExitThread_
< void(const Feedback &)>
< void(const WrappedResult &)>
< void()>
boost::signals2::signal < void(const Feedback &), optional_last_value < typename boost::function _traits< void(const Feedback &)>::result_type >, int, std ::less< int >, function< void (const Feedback &)>, typename extended _signature< function_traits< void(const Feedback &)>::arity, void(const Feedback &)>::function_type, boost::signals2::mutex >
boost::signals2::signal < void(const WrappedResult &), optional_last_value < typename boost::function _traits< void(const WrappedResult &)>::result_type >, int, std:: less< int >, function< void(const WrappedResult &)>, typename extended _signature< function_traits< void(const WrappedResult &)>::arity, void(const WrappedResult &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >
boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
smacc2::SmaccAsyncClient Behavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
-isShutdownRequested_
+isInitialized
-tfListener_
-listenerMutex_
-pose_
-tfBuffer_
#publishMessages
+frozenReferenceFrameTime
bool
shared_ptr< tf2_ros ::TransformListener >
mutex
PoseStamped
shared_ptr< tf2_ros ::Buffer >
cl_nav2z::CpPose
+ CpPose()
+ CpPose()
+ onInitialize()
+ update()
+ waitTransformUpdate()
+ toPoseMsg()
+ toPoseStampedMsg()
+ getYaw()
+ getX()
+ getY()
+ getZ()
+ setReferenceFrame()
+ getReferenceFrame()
+ getFrameId()
+ freezeReferenceFrame()
+ unfreezeReferenceFrame()
#robotPose_
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
-lastUpdate_
-periodDuration_
optional< rclcpp::Time >
optional< rclcpp::Duration >
vector< cl_nav2z::odom _tracker::CpOdomTracker ::PathInfo >
OdomTrackerStrategy
#pathInfos_
#strategy_
cl_nav2z::odom_tracker ::CpOdomTracker
+ CpOdomTracker()
+ processNewPose()
+ odomMessageCallback()
+ update()
+ setWorkingMode()
+ setPublishMessages()
+ pushPath()
+ pushPath()
+ popPath()
+ clearPath()
+ setStartPoint()
+ setStartPoint()
+ setCurrentMotionGoal()
+ setCurrentPathName()
+ getCurrentMotionGoal()
+ getPath()
+ logStateString()
+ updateParameters()
+ setOdomFrame()
# onInitialize()
# updateConfiguration()
# rtPublishPaths()
# updateRecordPath()
# updateClearPath()
# updateAggregatedStackPath()
# parametersCallback()
Path
WorkingMode
vector< nav_msgs::msg ::Path >
optional< geometry _msgs::msg::PoseStamped >
#aggregatedStackPathMsg_
#baseTrajectory_
#workingMode_
#pathStack_
#currentMotionGoal_
#odomSub_
#robotBasePathPub_
#robotBasePathStackedPub_
#robotPoseTimer_
#currentPathName_
#odomFrame_
#odomTopicName_
#param_callback_handle_
#clearAngularDistanceThreshold_
#clearPointDistanceThreshold_
#recordAngularDistanceThreshold_
#recordPointDistanceThreshold_
-poseFrameName_
-referenceFrame_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
#owner_
#m_mutex_
-m_mutex_
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
#components_
-orthogonal_
map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
-mutex_
-clientBehaviors_
#clients_
-stateMachine_
mutex
vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >
vector< std::shared _ptr< smacc2::ISmaccClient > >
#stateMachine_
-currentOrthogonal
-currentState
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
-stateMachine_
#eventGenerators_
#logger_
#stateReactors_
-signalDetector_
-smaccStateMachine_
#stateInfo_
#node_
vector< std::shared _ptr< smacc2::SmaccEventGenerator > >
shared_ptr< rclcpp ::Logger >
vector< std::shared _ptr< smacc2::StateReactor > >
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExitting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
recursive_mutex
StateMachineInternalAction
map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std ::pair< std::function < std::string()>, boost ::any > >
vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >
SharedPtr
SharedPtr
shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >
shared_ptr< smacc2 ::introspection::SmaccState Info >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >
int
vector< smacc2::introspection ::SmaccTransitionInfo >
vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >
type_info *
std::enable_shared _from_this< SmaccStateInfo >
map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >
map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >
string
SharedPtr
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#currentState_
#status_msg_
#orthogonals_
#nh_
#timer_
#stateMachineInfo_
#currentStateInfo_
+stateMachine_
+parentState_
+eventGeneratorsInfo
+depth_
+stateIndex_
+transitions_
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
+children_
+tid_
+staticBehaviorInfo
+stateReactorsInfo
+demangledStateName
+fullStateName
smacc2::introspection ::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
double
ExecutionModel
-statusPub_
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
atomic< int64_t >
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
-loop_rate_hz
-executionModel_
-updatableClients_
-updatableStateElements_
-lastState_
-end_
-initialized_
-rosInitialized_
-scheduler_
-processorHandle_
-signalDetectorThread_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()